# Lesson Notes

MODULE 2 | LESSON 2

# BEYOND MEAN-VARIANCE OPTIMIZATION

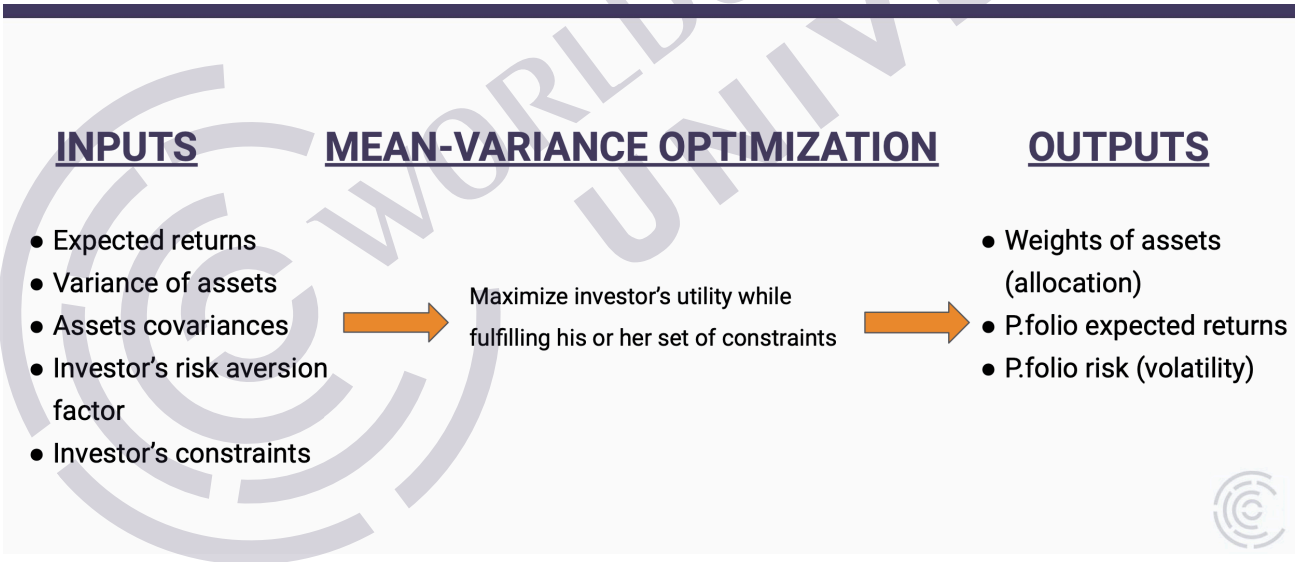
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| **Reading Time** | 150 minutes |
| **Prior Knowledge** | A sound understanding of the previous lesson, and an ability to manipulate matrices and using symbolic calculations (scipy library in Python) |
| **Keywords** | Coskewness, Cokurtosis, Kronecker products, General optimization (beyond quadratic programming), Symbolic calculations with Python. |

*In the lesson that precedes the current one, we look at how the inputs regarding the means and the covariance matrix of returns had a significant impact on the computation of the efficient frontier and the optimal portfolio itself. This lessons looks at how to extend the optimization program to make the problem of identifying the best portfolio more suitable for real life applications.*

## 1. Generalities

Optimizing a portfolio with respect to mean and variance is rather simplistic. As shown in Figure 1, the number of inputs considered is very limited.

**Figure 1. Mean-variance optimization and how it works.**




There are several criticisms of this technique. It does have the merit of being convenient, but it does not address several issues:

- quality of the inputs (which we discussed in the previous lesson);
- concentrated asset class allocation (also discussed already in Module 1);
- skewness and kurtosis (considered in this lesson);
- risk-diversification (asset classes could be driven by same factors, i.e., share some of the same sources of risk);
- ignores liabilities (sometimes investors need cash to pay for certain liabilities they have in the future);
- single period timeframe (it does not take into account cash flows over time or the serial correlation of returns among different asset classes).

In this lesson, we will look specifically at how to deal with skewness and kurtosis, and we will also consider the issue of liquidity. This will bring us to a multi-criteria objective function, sort of, where more than just expected returns and volatilities are of concern. We will also add additional criteria to our optimization problem.

We must be clear about one aspect: It is impossible to exhaust the topic that has been the subject of entire master's projects and doctoral theses; nevertheless, it is important for practitioners to have a thorough understanding of these issues.

## 2. Coskewness

Computing the skewness of a portfolio directly from, say, the daily returns or the monthly returns of the portfolio itself is enough to assess certain characteristics of the portfolio. However, if we wanted to assess the impact of different single assets on the overall skewness of a portfolio, we would need to introduce the concept of **coskewness**. We will spend a fair amount of time on coskewness, but it will allow us to move much quicker when dealing with cokurtosis in the next section.

Simply put, coskewness is the ***third standardized cross central moment***. In more practical terms, it provides a measure of how much three random variables change together (we know already that the covariance is a measure of how two random variables change together, instead). The formal definition for random variables is the following:

$$\gamma_{XYZ} = \frac{E[(X-\mu_X)\cdot(Y-\mu_Y)\cdot(Z-\mu_Z)]}{\sigma_X \cdot \sigma_Y \cdot \sigma_Z}$$

where $\mu_X, \mu_Y,$ and $\mu_Z$ are the expected values of the random variables $X, Y,$ and $Z,$ respectively. Notice that if $X = Y = Z,$ the formula above becomes:

$$\gamma_{XXX} = \frac{E[(X-\mu_X)^3]}{\sigma_X^3},$$

i.e., the usual measure of skewness for the case of an individual random variable. We use the symbol $\gamma$ because $s$ or $\sigma$ are already used for variances and covariances.

We will need to sweat it a bit and earn a deeper understanding of what coskewness is all about and how to represent it. For instance, we also need to deal with elements like

$$\gamma_{XXY} = \frac{E[(X-\mu_X)^2\cdot(Y-\mu_Y)]}{\sigma_X^2 \cdot \sigma_Y}$$

etc. In an attempt to simplify the notation as much as possible, we use $M_i, i = 1, 2, 3, 4$ to denote the $i$-th central moment matrix of returns. Thus, $M_2 = \Sigma,$ the variance-covariance matrix, which we know to contain moments like $E[X-\mu_X]^2, E[Y-\mu_Y]^2,$ and $E[X-\mu_X]E[Y-\mu_Y].$ Instead, $M_3$ is used to represent the coskewness matrix. Later, we will see that $M_4$ will be adopted as a symbol for the cokurtosis matrix.

The notation becomes quickly complicated. However, in the case of a $2$-asset portfolio we have

$$M_3 = \begin{pmatrix} \gamma_{111} & \gamma_{112} & \gamma_{121} & \gamma_{122} \\ \gamma_{211} & \gamma_{212} & \gamma_{221} & \gamma_{222} \end{pmatrix}.$$

We should notice at least three things about this matrix:

- It is a $2 \times 2^2 = 2 \times 4$ matrix.
- It contains only four distinct element since $\gamma_{112} = \gamma_{121} = \gamma_{211}, \gamma_{122} = \gamma_{212} = \gamma_{221}.$
- In general, in the case of an $m$-asset portfolio, the matrix $M$ is going to be a $m \times m^2$ matrix consisting of $m(m+1)(m+2)/6$ distinct elements.

**Example**. In the case of the $S\&P500$ index, $M_3$ would consist of $500 \cdot 500^2 = 125,000,000$ elements of which $20,968,500$ are distinct. So, computationally speaking, this is not a small task.

Why all this effort? Well, we need to be able to express the skewness of an $m$-asset portfolio as a function of the weights of each of the individual assets. If we are dealing with an $m$-asset portfolio and we describe it by the system of weights $w^T = (w_1, w_2, \ldots, w_m),$ the skewness of the overall portfolio has the following formula:

$$\text{skewness of portfolio } \equiv \gamma = w^T M_3 (w \otimes w).$$

The notation $w \otimes w$ is called a **tensor product** or a **Kronecker product** and it means the following:

$$w \otimes w = \begin{pmatrix} w_1 \cdot w_1 \\ w_1 \cdot w_2 \\ w_2 \cdot w_1 \\ w_2 \cdot w_2 \end{pmatrix}.$$

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There is nothing mysterious about this, and it is simply the consequence of how Kronecker products work. In general, if $a$ is a $m \times n$ matrix and $B$ is a $p \times q$ matrix, then

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \dots & a_{1n}B \\ a_{21}B & a_{22}B & \dots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \dots & a_{mn}B \end{pmatrix}$$

which is an $mp \times nq$ matrix. in our case $A = w$ a $n \times 1$ matrix and $B = w$ a $n \times 1$ matrix so that $w \otimes w$ is a $2 \cdot 2 \times 1 \cdot 1 = 4 \times 1$ matrix.

To get more acquainted with the notion, we compute the skewness of a $2$-asset portfolio using the formula provided:

$$\begin{aligned} \text{skewness}_p \equiv \gamma_p &= w^T M_3 (w \otimes w) \\ &= (w_1, w_2) \cdot \begin{pmatrix} \gamma_{111} & \gamma_{112} & \gamma_{211} & \gamma_{212} \\ \gamma_{121} & \gamma_{122} & \gamma_{221} & \gamma_{222} \end{pmatrix} \cdot \begin{pmatrix} w_1 \cdot w_1 \\ w_1 \cdot w_2 \\ w_2 \cdot w_1 \\ w_2 \cdot w_2 \end{pmatrix} \\ &= \gamma_{111} w_1^3 + (\gamma_{121} + \gamma_{112} + \gamma_{211}) w_1^2 w_2 + (\gamma_{122} + \gamma_{221} + \gamma_{212}) w_1 w_2^2 + \gamma_{222} w_2^3. \end{aligned}$$

We could simplify it even further by recalling that the terms inside each group of parentheses are all the same. We are starting to see how the skewness of a portfolio relates to the weights and the coskewness of individual assets.

Luckily, we do not need to do this by hand. Some students may feel intimidated by the use of tensor notation/Kronecker products, but most relevant software has Kronecker products built in. For instance, Python has the function **numpy.kron** while R has the product $\%x\%$, instead.

**Example (Kronecker products in Python)**. Say that we wanted to perform the Kronecker product of two vectors $a^T = (1, 2, 3)$ and $b^T = (0, 1, 1)$. We can have Python do it for us as follows:

```
import numpy as np
a = np.array([[1],[2],[3]])
b = np.array([[0],[1],[1]])
result = np.kron(a, b)
print(" a kronecker b = ")
print(result)

a kronecker b =
[[0]
[1]
[1]
[0]
[2]
[2]
[0]
[3]
[3]]
```

For the Kronecker product of two or more matrices, things are not really different. But there would be potentially a large output. For instance,

```
A = [[1, 2], [0,2]]
B = [[1,1], [1,-1]]
C = np.kron(A, B)
print(C)
```

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```
[[ 1  1  2  2]
[ 1 -1  2 -2]
[ 0  0  2  2]
[ 0  0  2 -2]]
```

**Example. (Computing skewness of a portfolio from coskewness of individual assets)**. A portfolio consists of two assets and the weights of each asset are $w^T = (0.4, 0.6)$. The co-skewness matrix is the following:

$$M_3 = \begin{pmatrix} 0.2 & 0.1 & 0.1 & 0.3 \\ 0.1 & 0.3 & 0.3 & 0.5 \end{pmatrix}.$$

Find the skewness of the portfolio.

The formula is the one given earlier, i.e.:

$$\text{skewness}_p = w^T M_3 (w \otimes w).$$

This could still be done by hand using the formula shown earlier. However, we can also use Python to do it for us.

```
import numpy as np
w = np.array([[0.4],[0.6]])
M =np.array([[0.2, 0.1, 0.1, 0.3],[0.1,0.3,0.3,0.5]])
Sk = np.matmul(np.matmul(np.transpose(w), M), np.kron(w,w))
print(Sk)
[[0.2792]]
```

i.e. the skewness of the portfolio is 0.2792.

Before we move forward, it is useful to think about a few relevant takeaways:

- **Positive coskewness** means that the three stocks will tend to undergo extreme deviations at the same time (but notice that they could be all three movements in the positive direction or 2 in the negative direction and 1 in the positive direction)
- **Negative coskewness** also means that three stocks will tend to undero extreeme movements at the same time and either all 3 moves in the negative direction or one moves in the negative direction and two in the positive direction.

For a portfolio it is difficult to quantify what the actual practical implications are. Of course we like when positive coskewness means that all three stocks will move in the positive direction, but we like it less when the positive coskewness is the consequence of two stocks undergoing large negative movements and only one exhibits a large positive movement. Thomas Kim in a paper quoted in the reference looked into this problem and found that, while high coskewness stocks tend to increase the skewness of the portfolio (a good thing), the number of assets had a larger impact on the skewness of the portfolio than the individual assets' coskewnesses and a large number of assets tends to eliminate the impact of high positive coskewness assets. But it is not a conclusive study.

Can we represent coskewness graphically? Definitely not as easily or effectively as the covariance. However, the next example, taken from Wikipedia, supplies all the calculations and a couple of plots to it (see *coskewness from Wikipedia*). The gist of this example can be summarized like this: *We can start with two random variables, which individually have no skewness. However, unless the two random variables are independent, their sum can have either negative or positive skewness (depending on the correlation).*

Students who are interested can find the details below. Some mathematical sophistication is required, but ambitious quants should definitely welcome the challenge.

**Example (Skewness and coskewness. OPTIONAL)**. Let $X$ be a standard normal random variable and let $Y$ be another random variable defined as follows:

$$Y \sim \begin{cases} X & \text{if } X < 0; \\ T & \text{if } X \geq 0; \end{cases}$$

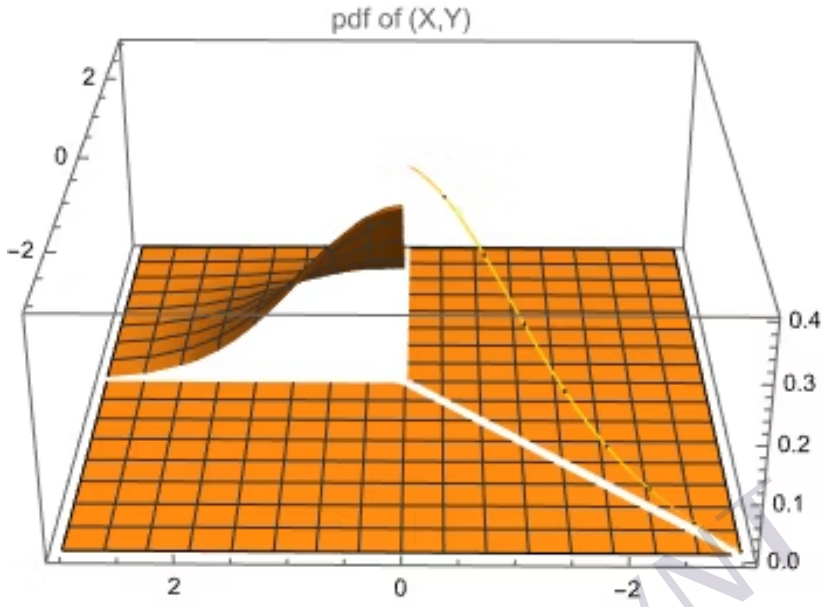
where $T$ is a half-normal random variable defined on the positive half of the real line and independent of $X$.

Then, the joint pdf of $X$ and $Y$ is given by

$$f_{X,Y}(x,y) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}\left(\delta(x-y)\cdot 1_{x<0} + 2\frac{e^{-y^2/2}}{\sqrt{2\pi}}\cdot 1_{x\geq 0}\cdot 1_{y\geq 0}\right)$$

where $\delta(x)$ is the Dirac delta function. The picture of the joint density is shown in Figure 2 below. Note that the visual effect along the diagonal in the negative quadrant has been magnified so to be evident in the picture.

**Figure 2: The Joint pdf of $X$ and $Y$.**




Now, both $X$ and $Y$ are marginally $\mathcal{N}_{0,1}$ random variables. For instance:

$$f_Y(y) = \int_{-\infty}^{0} \frac{e^{-x^2/2}}{\sqrt{2\pi}}\delta(x-y)\,dx + \int_0^{\infty} \frac{e^{-x^2/2}}{\sqrt{2\pi}} 2\frac{e^{-y^2/2}}{\sqrt{2\pi}}\,dx.$$

For the first integral, we need to use the property of the Dirac delta function, i.e.,

$$\int_{\infty}^{\infty} f(y)\delta(x-y)\,dy = f(x).$$

In our case, this becomes:

$$\int_{-\infty}^{0} \frac{e^{-x^2/2}}{\sqrt{2\pi}}\delta(x-y)\,dx = \frac{e^{-y^2/2}}{\sqrt{2\pi}}\cdot 1_{\{y<0\}}.$$

The other integral is easier to handle and yields

$$\int_0^{\infty} \frac{e^{-x^2/2}}{\sqrt{2\pi}} 2\frac{e^{-y^2/2}}{\sqrt{2\pi}}\,dx = \frac{e^{-y^2/2}}{\sqrt{2\pi}}\cdot 1_{\{y\geq 0\}}$$

and which proves that $Y$ is marginally a $\mathcal{N}_{0,1}$ random variable as claimed.

As both $X$ and $Y$ are $\mathcal{N}_{0,1}$ random variables, neither has any skewness.

However, it is clear that $X$ and $Y$ are not independent (why?), and their sum turns out to be negatively skewed. This can be shown in two ways (which are great exercises in master's level statistics).

The first method uses a change of variables: Let $U = X + Y$ and $V = X$. Then $X = V, Y = U - V$, and

$$\begin{aligned} f_U(u) &= \int_{-\infty}^{0} \frac{e^{-v^2/2}}{\sqrt{2\pi}}\delta(2v-u)\cdot 1_{\{u<0\}}\,dv + 2\int_0^{\infty} \frac{e^{-v^2/2}e^{-(u-v)^2/2}}{2\pi}\cdot 1_{\{u\geq v\}}\,dv \\ &= \int_{-\infty}^{0} \frac{e^{-v^2/2}}{\sqrt{2\pi}}\frac{1}{2}\delta\left(v-\tfrac{u}{2}\right)\cdot 1_{\{u<0\}}\,dv + \frac{e^{-u^2/2}}{\pi}\int_0^{\infty} e^{-v^2/2+uv}\cdot 1_{\{u\geq v\}}\,dv. \end{aligned}$$

Now, from the properties of the delta function, we have that the first integral is equal to

$$\frac{e^{-(u/2)^2/2}}{2\sqrt{2\pi}}\cdot 1_{u<0}.$$

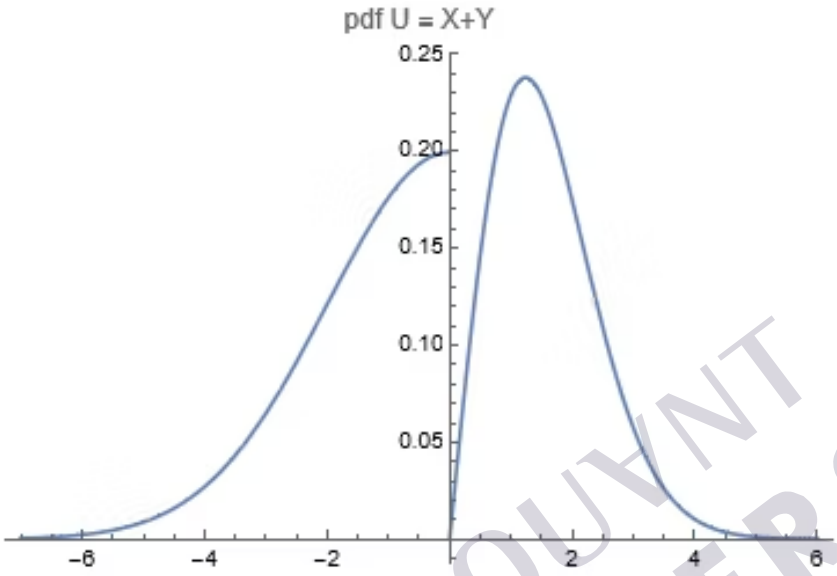
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The second integral, if we add and subtract $u^2/4$ to the power of the exponential function, after a few algebraic manipulations, should be

$$\frac{e^{-u^2/4}}{2\sqrt{\pi}}\int_0^u \frac{2}{\sqrt{\pi}}e^{-(v-u/2)^2}\,dv = \frac{e^{-u^2/4}}{2\sqrt{\pi}}\int_{-u/2}^{u/2}\frac{2}{\sqrt{\pi}}e^{-(w)^2}\,dw$$
$$= \frac{e^{-u^2/4}}{2\sqrt{\pi}}\cdot 2\text{erf}(u/2)\cdot 1_{\{u\geq 0\}} = \frac{e^{-u^2/4}}{\sqrt{\pi}}\cdot \text{erf}(u/2)\cdot 1_{\{u\geq 0\}}.$$

The pdf of the sum of $X$ and $Y$ is shown in Figure 3 and it displays some negative skewness, despite neither $X$ nor $Y$ having any skewness of their own. Dependence is what is causing it.

**Figure 3: The Distribution of $X+Y$.**




A second way to establish this result is to use the definition of coskewness directly. Simple algebra shows that

$$\gamma_{X+Y} = \frac{1}{\sigma^3_{X+Y}}\cdot\left(\sigma^3_X\gamma_{XXX} + 3\sigma^2_X\sigma_Y\gamma_{XXY} + 3\sigma_X\sigma^2_Y\gamma_{XYY} - \sigma^3_Y\gamma_{YYY}\right).$$

We know already that $\gamma_{XXX} = \gamma_{YYY} = 0$ because the two marginal distributions are standard normal. We also know that $\sigma^2_X = \sigma^2_Y = 1$ (both marginal distributions are $N(0,1)$). Furthermore, since $\mu_X = 0$ and $\mu_Y = 0$,

$$\gamma_{XXY} = \int_{-\infty}^0\int_{-\infty}^0 x^2 y\frac{e^{-x^2/2}}{\sqrt{2\pi}}\delta(x-y)\,dxdy + \int_0^\infty\int_0^\infty 2\frac{e^{-x^2/2}}{\sqrt{2\pi}}\frac{e^{-y^2/2}}{\sqrt{2\pi}}\,dxdy$$
$$= \int_{-\infty}^0 x^2\frac{e^{-x^2/2}}{\sqrt{2\pi}}\,dx\int_{-\infty}^0 y\delta(x-y)\,dy + \int_0^\infty 2x^2\frac{e^{-x^2/2}}{\sqrt{2\pi}}\,dx\int_0^\infty\frac{e^{-y^2/2}}{\sqrt{2\pi}},dy$$
$$= \int_{-\infty}^0 x^3\frac{e^{-x^2/2}}{\sqrt{2\pi}}\,dx + \int_0^\infty 2x^2\frac{e^{-x^2/2}}{\sqrt{2\pi}}\,dx\int_0^\infty\frac{e^{-y^2/2}}{\sqrt{2\pi}}\,dy,$$

using once more the property of the Dirac delta function for the first integral. At this point, the first integral yields

$$\int_{-\infty}^0 x^3\tfrac{e^{-x^2/2}}{\sqrt{2\pi}}\,dx = -\int_{-\infty}^0\tfrac{x^2}{\sqrt{2\pi}}\tfrac{de^{-x^2/2}}{dx}\,dx$$

which can be integrated by parts and yields

$$\int_{-\infty}^0 x^3\frac{e^{-x^2/2}}{\sqrt{2\pi}}\,dx = -\frac{\sqrt{2}}{\sqrt{\pi}}.$$

The second integral is separable, and, using the moments of the half-normal distribution, we find that

$$\int_0^\infty 2x^2\frac{e^{-x^2/2}}{\sqrt{2\pi}}\,dx\int_0^\infty\frac{e^{-y^2/2}}{\sqrt{2\pi}}\,dy = \frac{1}{\sqrt{2\pi}}.$$

Thus, putting everything together, we have shown that $\gamma_{XXY} = -\frac{1}{\sqrt{2\pi}}$. Similarly, one shows that $\gamma_{XYY} = -\frac{1}{\sqrt{2\pi}}$, as well. Furthermore, we can show that $Cov(X,Y) = \frac{\pi+2}{2\pi}$ and from that, $\sigma_{X+Y} = \sqrt{\frac{3\pi+2}{\pi}}$. Finally:

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$$\gamma_{X+Y} = -\frac{3\sqrt{2}\pi}{(2+3\pi)^{3/2}} \approx -0.345.$$

In other words, we have produced an example where, starting with two normal random variables with zero skewness, their sum exhibits instead negative skewness due to the nature of the correlation between the two original random variables. That is why coskewness is relevant!

Other more useful discussions and some visuals about the types of coskewness and cokurtosis can be found in the following reading: Least Quartic Regression Criterion to Evaluate Systematic Risk in the Presence of Co-Skewness and Co-Kurtosis

We have learned a lot about coskweness so far and how it relates to skewness of a portfolio starting from the individual assets. However, we also saw that the coskweness matrix needed, $M_3$, can be really a big matrix to compute even if many elements are the same. Thus, $M_3$ has to be estimated from the returns data of the individual assets over the reference period.

How do we go from the theoretical formula for coskewness for random variables to one that uses the sample data? For instance:

$$\widehat{\gamma}_{112} = \frac{\sum_{t=1}^{N}(r_{1t} - \bar{r}_1)(r_{1t} - \bar{r}_1)(r_{2t} - \bar{r}_2)}{(N-1)s_1 s_1 s_2} = \frac{\sum_{t=1}^{N}(r_{1t} - \bar{r}_1)^2(r_{2t} - \bar{r}_2)}{(N-1)s_1^2 s_2},$$

etc.

## 3. Cokurtosis

The previous section was quite long because we discussed the issues related to the concept of coskewness. The same basic ideas apply to the notion of cokurtosis. **Cokurtosis** deals with the standardized $4$-th central moments of random variables. Unlike covariance, which involves two random variables at a time, or coskewness, which involves three random variables at a time, cokurtosis uses four variables at the same time.

We will use the symbol $M4$ for the kurtosis matrix.

The concept of cokurtosis is important for applications in portfolio theory because if two random variables or assets exhibit a high level of cokurtosis, then they will move together undergoing either large positive or large negative deviations during the assigned reference period. This obviously has implications for when one uses VaR to assess the risk of portfolios since coskweness and cokurtosis would tend to produce distributions with heavy tails and/or that are skewed.

The main practical complication stems from the fact that when using fourth moments, there are many of them.

In the case of four random variables, $W, X, Y,$ and $Z,$ then cokurtosis is defined by

$$K_{WXYZ} = \frac{E[(W - \mu_w) \cdot (X - \mu_X) \cdot (Y - \mu_Y) \cdot (Z - \mu_Z)]}{\sigma_W \sigma_X \sigma_Y \sigma_Z}$$

and all its possible variations like for example

$$K_{XXYZ} = \frac{E[(X - \mu_X)^2 \cdot (Y - \mu_Y) \cdot (Z - \mu_Z)]}{\sigma_X^2 \sigma_Y \sigma_Z},$$

etc. In particular when $W = Y = Z = X$ we have the traditional definition of kurtosis for a random variable, that is:

$$K_X = \frac{E[(X - \mu_X)^4]}{\sigma_X^4}.$$

For applications, it is more important to use the sample formula instead of the theoretical one for random variables/populations as we will be using samples of returns from assets. To this purpose, if we consider the returns for four assets in the portfolio, say $r_h, r_i, r_j, r_k,$ over a period of, say $N$- days, then the sample cokurtosis for the returns of these 4 assets is given by:

$$\widehat{K}_{hijk} = \frac{N(N+1)(N-1)}{(N-2)(N-3)} \cdot \frac{\sum_{t=1}^{N}(r_{ht} - \bar{r}_h)(r_{it} - \bar{r}_i)(r_{jt} - \bar{r}_j)(r_{kt} - \bar{r}_k)}{(\sum_{t=1}^{N}(r_{ht} - \bar{r}_h)^2)^{1/2} \cdot (\sum_{t=1}^{N}(r_{it} - \bar{r}_i)^2)^{1/2} \cdot (\sum_{t=1}^{N}(r_{jt} - \bar{r}_j)^2)^{1/2} \cdot (\sum_{t=1}^{N}(r_{kt} - \bar{r}_k)^2)^{1/2}}$$
$$\approx \frac{N \cdot \sum_{t=1}^{N}(r_{ht} - \bar{r}_h)(r_{it} - \bar{r}_i)(r_{jt} - \bar{r}_j)(r_{kt} - \bar{r}_k)}{(\sum_{t=1}^{N}(r_{ht} - \bar{r}_h)^2)^{1/2} \cdot (\sum_{t=1}^{N}(r_{it} - \bar{r}_i)^2)^{1/2} \cdot (\sum_{t=1}^{N}(r_{jt} - \bar{r}_j)^2)^{1/2} \cdot (\sum_{t=1}^{N}(r_{kt} - \bar{r}_k)^2)^{1/2}}$$

if $N$ is large. The last formula can be rewritten in a more compact manner as:

$$\frac{1}{N-1}\frac{\sum_{t=1}^{N}(r_{ht}-\bar{r}_h)(r_{it}-\bar{r}_i)(r_{jt}-\bar{r}_j)(r_{kt}-\bar{r}_k)}{s_h s_i s_j s_k} \quad \text{if N is large}$$

The cokurtosis matrix,$M_4$, is now very large. For just two assets, it looks like this:

$$M_4 = \begin{pmatrix} K_{1111} & K_{1112} & K_{1121} & K_{1122} & K_{1211} & K_{1212} & K_{1221} & K_{1222} \\ K_{2111} & K_{2112} & K_{2121} & K_{2122} & K_{2211} & K_{2212} & K_{2221} & K_{2222} \end{pmatrix}.$$

Most of these elements are the same, but we should notice that even for a two-asset problem,$M_4$ is a$2 \times 2^3 = 8$ matrix. In case we were working with $m$ assets, the number of elements in $M_4$ would become a $m \times m^3$ matrix, which can be quite large in the case of index portfolios. For the S&P 500, it would be a matrix of dimensions $500 \times 500^3 = 500 \times 125,000,000$! However, one could check using combinatorics (try counting the number of unordered arrangements of size r from m object while sampling with replacement), that the number of distinct entries is only:

$$1 + \binom{m-1}{1} + \binom{m}{2} + \binom{m+1}{3} + \binom{m+2}{4}.$$

The kurtosis of a portfolio with weights $w^T = (w_1, w_2, \ldots, w_m$ is obtained as:

$$\text{kurtosis of portfolio } = w^T M_4 (w \otimes w \otimes w).$$

## 4. Optimal Portfolios Adding Skewness and Kurtosis

Let's start by looking at a couple of plots involving our portfolio of eight assets that we have used many times already in previous lessons.

Figure 4 shows 10,000 simulated portfolios from eight assets. The portfolios close to the efficient frontier do not have all the same skewness, and negative skewness between -0.1 and -0.2 appears the most characteristic value.

**Figure 4: 10,000 Simulated Portfolios and Their Skewness**

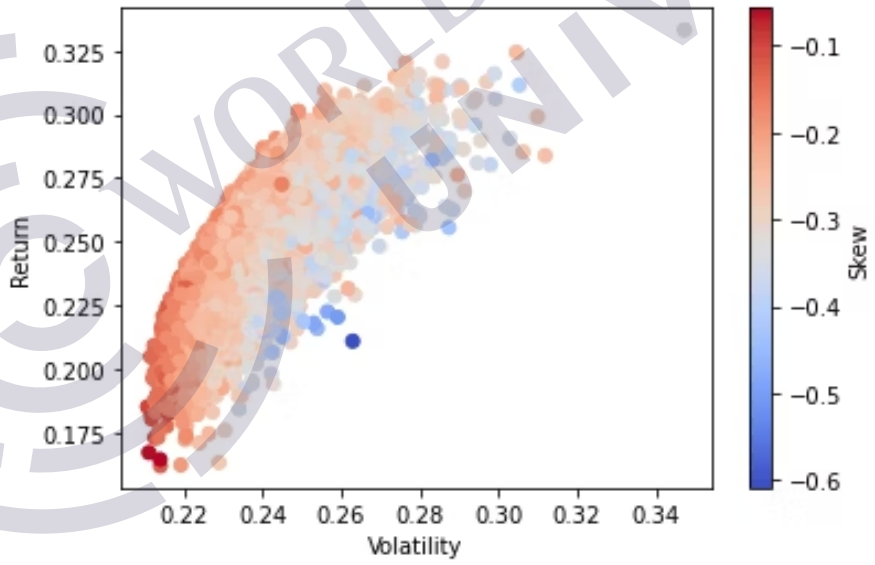
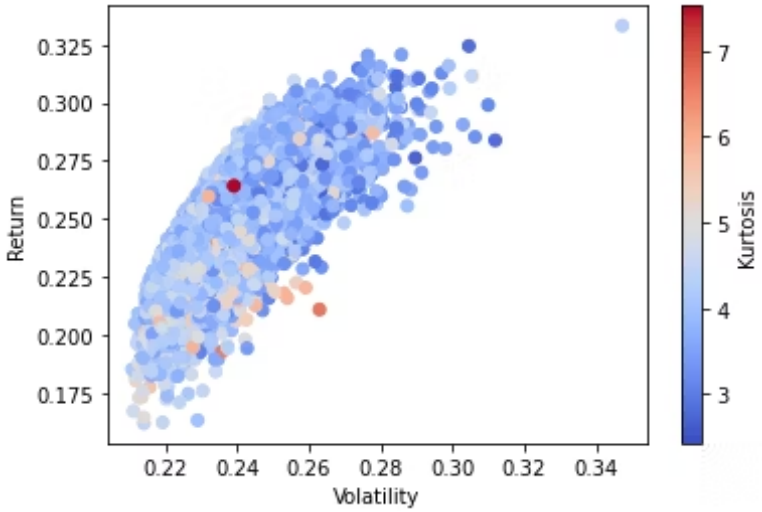



Figure 5 below also shows the results of the same 10,000 simulated portfolios, but in this case, we look at the kurtosis of those portfolios. The majority of these portfolios have kurtosis between 4 and 5, i.e., heavier tails than in the case of a normal distribution.

**Figure 5: 10,000 Simulated Portfolios and Their Kurtosis**

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Considering the aspects highlighted in the two figures immediately above, it makes sense that to protect portfolios from risk, while trying to optimize the risk-return trade-off, skewness and kurtosis should be introduced in the optimization program. Hence, our optimization program becomes:

$$\max_{w} \alpha_1 w^T M_1 + \alpha_2 w^T M_2 w + \alpha_3 w^T M_3 (w \otimes w) + \alpha_4 w^T M_4 (w \otimes w \otimes w)$$
$$\text{subject to} \quad w^T \iota = 1.$$

Note that we wrote the program using the matrices/arrays of moments to make it clear that we are now using additional moments compared with what we did in earlier lessons.

$\alpha_1, \alpha_2, \alpha_3,$ and $\alpha_4$ are parameters that control how important that particular moment is. While the magnitude of those parameters can potentially change among individuals, it seems quite realistic to assume that $\alpha_1 > 0$ ,$\alpha_2 < 0$ (economic agents would prefer to avoid volatility/risk as much as possible, but with differing degrees among different individuals),$\alpha_3 > 0,$ and $\alpha_4 < 0.$ The rationales are that:

- $\alpha_1 > 0,$ since economic agents want a high expected return;
- $\alpha_2 < 0,$ as agents would prefer to avoid volatility/risk as much as possible, but with differing degrees among different individuals;
- $\alpha_3 > 0,$ because economic agents welcome positive skewness but dislike negative skewness;
- $\alpha_4 < 0,$ because higher kurtosis has the ability to cause large losses (as well as large gains). The difference among individuals in fearing kurtosis is modeled via the size of the parameter as is the case for the other three as well.

Needless to say, we can add many more constraints to our optimization program. Now, it should also be completely clear why we devoted a considerable amount of time to the notions of coskewness and cokurtosis: We needed to express those statistics as a function of the weights of the individual assets that are/could be considered in forming our portfolio. Now, we have an optimization problem.

## 5. Liquidity Constraints

Most would agree that liquidity can be defined as the efficiency or ease with which an asset or security can be converted into ready cash without affecting its market price. It is a very desirable feature, and in addition, liquid assets imply an awareness that one can buy or sell assets within fair price levels. Liquidity can be characterized in more than one way.

Securities that are fungible, i.e. mutually interchangeable, are more likely to be liquid than securities, for which no two are exactly alike. For instance, homes are not necessarily fungible in general (due to different locations, sizes, etc.). Expensive arts are likely illiquid assets as one has to find suitable buyers with both the interest and the money to purchase the assets. At the opposite extreme, the stock market is characterized by higher market liquidity. In particular, whenever an exchange has a high volume of trade that is not dominated by selling, the price a buyer offers per share (bid price) and the price the seller is willing to accept (ask price) will be fairly close to each other.

There is more than one way to measure the liquidity of an asset/investment: bid-ask spread, as we just mentioned in the previous paragraph; volume-related liquidity measures such as the volume or quantity of shares traded or the number of transactions/orders per unit of time; or turnover ratio, that is the proportion of shares of a stock traded during a specific time reference (say one month) out of the outstanding number of shares of that particular security. Often, the type of assets dictate what measure one can use. We will adopt a turnover ratio as a measure of liquidity, but the student should be aware that we are

not committed to a particular measure of liquidity; we are just concentrating on how to use liquidity in our portfolio selection process.

Once we have calculated the liquidity measure of each asset/stock, say $L_i, i = 1, 2, \ldots, m,$ the higher the number $L_i$ is, the more liquid an asset $i$ would be considered. For a portfolio, we could define liquidity as

$$L_{p.folio} = \sum_{i=1}^{m} w_i L_i \quad \text{or} \quad L_{p.folio} = w^T L,$$

where $L^T = (L_1, \ldots, L_m).$

This seems like a plausible measure of liquidity for a portfolio, and we can conjecture that the first moment of liquidity (mean) is important in portfolio optimization. However, there is nothing preventing us from using higher moments for liquidity as we did for returns. In fact, the individual liquidity of certain assets may **co-move**, i.e., in certain periods, all assets can trade very liquid and in others, many assets' volumes could dry up. It is also possible that when certain assets are very liquid, others see very limited trading. Since liquidity can drive returns, liquidity could enter as an element of our objective function. In addition, if the co-movement of liquidity does not follow patterns similar to those of returns, it makes sense to consider higher moments of liquidity in our objective function as well.

This leads us to expand our optimization problem to look like this:

$$\begin{aligned} \max_{w} \alpha_1 w^T M_1 + \alpha_2 w^T M_2 w + \alpha_3 w^T M_3 (w \otimes w) + \alpha_4 w^T M_4 (w \otimes w \otimes w) + \\ \beta_1 w^T M_1^L + \beta_2 w^T M_2^L w + \beta_3 w^T M_3^L (w \otimes w) + \beta_4 w^T M_4^L (w \otimes w \otimes w) \\ \text{subject to} \quad w^T \iota = 1, \end{aligned}$$

with $M_i^L, i = 1, 2, 3, 4$ representing the central moments for liquidity instead of returns and $\beta_i, i = 1, 2, 3, 4$ representing a specific investor's preferences. For instance, some investors could prefer higher moments while others may be uneasy and penalize the higher moments.

Also, we could rewrite the problem above as

$$\begin{aligned} \max_w \lambda_R \big(\alpha_1 w^T M_1 + \alpha_2 w^T M_2 w + \alpha_3 w^T M_3 (w \otimes w) + \alpha_4 w^T M_4 (w \otimes w \otimes w)\big) + \\ \lambda_L \big(\beta_1 w^T M_1^L + \beta_2 w^T M_2^L w + \beta_3 w^T M_3^L (w \otimes w) + \beta_4 w^T M_4^L (w \otimes w \otimes w)\big) \\ \text{subject to} \quad w^T \iota = 1, \end{aligned}$$

where the parameters $\lambda_R$ and $\lambda_L$ are modeling specific preferences for returns and/or liquidity.

This has led us to consider multiple criteria in selecting an optimal portfolio, and we can definitely add many more constraints. Hence, we believe that students who followed the exposition so far are now in a good place to appreciate several subtle nuances that come with portfolio optimization theory.

## 6. General Optimization Techniques

A problem like

$$\begin{aligned} \max_{w} \alpha_1 w^T M_1 + \alpha_2 w^T M_2 w + \alpha_3 w^T M_3 (w \otimes w) + \alpha_4 w^T M_4 (w \otimes w \otimes w) \\ \text{subject to} \quad w^T \iota = 1. \end{aligned}$$

is not an instance of quadratic programming (QP) anymore. We are still dealing with a polynomial objective function, but the variables $w_1, \ldots, w_m$ appear with powers higher than two. This means that one has to use more powerful techniques that can handle this type of problem, and we have to be aware that nonlinear constrained optimization is not an easy task in general. In fact, it could be a topic for an entire class on its own. Python provides several packages to solve these problems. We have neither the time nor space to investigate this topic in full, but we will look at how to deal with problems like the one at hand.

Specifically we will use the **minimize** package from the **scipy.optmize** library with the *SLSQP optimizer* (an optimizer based on a sequential least squares programming algorithm), and we will show how to use it by solving a specific problem that incorporates all relevant features.

$$\begin{aligned} &\min_{w_1, w_2, w_3, w_4} w_1 \cdot w_4 \cdot (w_1 + w_2 + w_3) + w_4 \ \ \text{(objective function)} \\ &\text{s.t.} \ \ w_1 w_2 w_3 w_4 \geq 25 \ \ \text{(inequality constraint)} \\ &\qquad w_1^2 + w_2^2 + w_3^2 + w_4^2 = 45 \ \ \text{(equality constraint)} \\ &\qquad 1 \leq w_1, w_2, w_3, w_4 \leq 5 \ \ \text{(bounds for the variables)} \\ &\qquad w_0 = (1, 6, 3, 2) \ \ \text{(initial guess for the solutions)} \end{aligned}$$

The Python code to solve this problem is the following:

```
import numpy as np
from scipy.optimize import minimize  # load the package minimize

def objective(w):  # define the objective function to minimize.
   return w[0]*w[3]*(w[0]+w[1]+w[2])+w[3]

def constraint1(w):   # define the typical inequality constraint
   return w[0]*w[1]*w[2]*w[3]-25.0

def constraint2(w):   # define the prototypical equality constraint.
   return w[0]**2 + w[1]**2 + w[2]**2 + w[3]**2 - 45.0

# initial weights
w0 = (1,6,3,2)

# show initial objective using the initial solution
print('Initial Objective: '+ str(objective(w0)))

# optimize
b = (1.0,5.0)
bnds = (b, b, b, b)   # define the interval bounds for each of the variables.
con1 = {'type': 'ineq', 'fun': constraint1}  # tell the program the type of constr.
con2 = {'type': 'eq', 'fun': constraint2}   # tell the program the type of constr.
cons = ([con1,con2])    # this is the list of constraints the program will receive
solution = minimize(objective,w0,method='SLSQP', bounds=bnds,constraints=cons)
w = solution.x

# show final objective
print('Final Objective: ' + str(objective(w)))

# print solution
print('Solution')
print('w1 = ' + str(w[0]))
print('w2 = ' + str(w[1]))
print('w3 = ' + str(w[2]))
print('w4 = ' + str(w[3]))
```

If we want to maximize a function, say $f(w_1, w_2, w_3)$, we will do so by minimizing $-f(w_1, w_2, w_3)$. Note the particular form of the inequality constraint. If we wanted $\leq$ instead of $\geq$, again, we can do that just by multiplying both sides of the inequality by $-1$.

Finally, one caveat: Non-linear constrained optimization is complex and the algorithm may not find the optimal global solution and could instead end up on a local solution. Sometimes, changing the initial condition could tell us whether that is the case or not.

Now we go back to our portfolio optimization problem, and this time, we add skewness and kurtosis to the mean and variance. The first task is to be able to write the objective function. To keep things reasonably simple, we are going to consider only three stocks and not eight as before. The objective function consists of four parts:

```
import sympy as sy
import numpy as np
import pandas as pd
import math
import yfinance as yf
from datetime import datetime
import matplotlib.pyplot as plt
import seaborn as sns
from pandas_datareader import data

### define the stocks from which we want the data (via tickers) ###
#assets = ["AAPL", "MSFT","AMZN","GOOGL","META","TSLA","AMGN", "V"]
assets_list = ["AAPL", "MSFT", "AMZN"] # we worked with a reduced dataset to simplify
# Define the start date
start_date = "2015-01-2"
end_date = "2022-01-15"
# Today's date
#today = datetime.today().strftime('%Y-%m-%d')
# Create an empty dataframe
df_prices = pd.DataFrame()
# Get the  "Adj Close" data from Yahoo Finance
def YahooData(dataframe,assets_list,start_date,end_date):
    for i in assets_list:
        dataframe[i] = yf.download(i, start=start_date, end=end_date)['Adj Close']
    return dataframe
df1 = YahooData(df_prices,assets, start_date,end_date)
# We transform the data to it´s logarithmic returns
df = np.log(df1).diff()
# Drop the first row because we loose information we the logarithmic return.
df = df.dropna()

# compute coskewness matrix

num = len(df.columns)
numc = len(df)
M3 = np.zeros(shape = (num, num**2))
v = df.values
means = v.mean(0,keepdims=True)
v1 = (v-means).T

for i in range(num):
    for j in range(num):
        for k in range(num):
            u = 0.0
            for t in  range(numc):
                u=u+v1[i,t]*v1[j,t]*v1[k,t]

            M3[i, j*num+k] = u/float((len(df)-1)\
            *df.iloc[:,i].std()*df.iloc[:,j].std()*df.iloc[:,k].std())
```

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```
M3

# Compute the cokurtosis matrix

num = len(df.columns)
numc = len(df)
M4 = np.zeros(shape = (num, num**3))
v = df.values
means = v.mean(0,keepdims=True)
v1 = (v-means).T
for i in range(num):
    for j in range(num):
        for h in range(num):
            for k in range(num):
                u = 0.0
                for t in  range(numc):
                    u=u+v1[i,t]*v1[j,t]*v1[h,t]*v1[k,t]

                M4[i, j*num**2+ h*num+k] = u/float((len(df)-1)\
                *df.iloc[:,i].std()*df.iloc[:,j].std()*df.iloc[:,h].std()\
                *df.iloc[:,k].std())

M4

w1 = sy.Symbol('w1')
w2 = sy.Symbol('w2')
w3 = sy.Symbol('w3')
w =[w1,w2,w3]

M1 = np.round(v.mean(0,keepdims=True),6)
# define mean of portfolio
meanP = sy.simplify(np.dot(w, M1.T))

M2 = df.cov()

# define variance of portfolio.
varP = np.dot(np.dot(w, M2), w)
varP= sy.simplify(varP)

h = np.dot(w, M3)
k =np.kron(w,w)
# define skewness of portfolio
skewP =sy.simplify(np.dot(h,k))

p= np.kron(w, w)
q =np.kron(w, p)
r = np.dot(w, M4)
```

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```
# define kurtosis of portfolio
kurtP = sy.simplify(np.dot(r, q))
```

Then in our example this code produced the following results (students who run this code will find similar formulae with different numbers)

$$meanP = 0.001072 \times w_1 + 0.001099 \times w_2 + 0.001288 \times w_3$$

$$varP = 0.000335w_1^2 + 0.000428w_1w_2 + 0.000392w_1w_3 + 0.00029w_2^2 + 0.000422w_2w_3 + 0.000369w_3^2$$

$$\begin{aligned} skewP = {} & 0.27365w_1^3 - 1.11516w_1^2w_2 - 0.56307w_1^2w_3 - 0.9627w_1w_2^2 - 1.04442w_1w_2w_3 \\ & - 0.46659w_1w_3^2 - 0.2243w_2^3 - 0.02658w_2^2w_3 + 0.16305w_2w_3^2 + 0.52807w_3^3 \end{aligned}$$

and

$$\begin{aligned} kurtP = {} & 9.36158w_1^4 + 30.61548w_1^3w_2 + 17.86736w_1^3w_3 + 50.36082w_1^2w_2^2 + 51.23004w_1^2w_2w_3 \\ & + 19.875w_1^2w_3^2 + 38.88744w_1w_2^3 + 57.08256w_1w_2^2w_3 + 40.94832w_1w_2w_3^2 \\ & + 11.10984w_1w_3^3 + 13.61024w_2^4 + 26.82224w_2^3w_3 + 31.76712w_2^2w_3^2 + 17.96152w_2w_3^3 + 9.70931w_3^4 \end{aligned}$$

Now, we can set up the problem like in the case of the optimization one. The objective function will be quite complex, but not particularly difficult.

```
alpha1 = -1   # note! The signs have been changed to switch the
alpha2 = +1/2 # maximization program to a minimization one.
alpha3 = -1
alpha4 = 1
def mean(w):
return alpha1*(0.001072*w[0] + 0.001099*w[1] + 0.001288*w[2])

def var(w):
return alpha2*(0.000335*w[0]**2 + 0.000428*w[0]*w[1] + 0.000392*w[0]*w[2]+ 0.00029*w[

def skew(w):
return alpha3*(0.27365*w[0]**3 - 1.11516*w[0]**2*w[1] - 0.56307*w[0]**2*w[2]
- 0.9627*w[0]*w[1]**2 - 1.04442*w[0]*w[1]*w[2] - 0.46659*w[0]*w[2]**2
 - 0.2243*w[1]**3 - 0.02658*w[1]**2*w[2] + 0.16305*w[1]*w[2]**2 + 0.52807*w[2]**3)

def kurt(w):
 return alpha4*(9.36158*w[0]**4 + 30.61548*w[0]**3*w[1] + 17.86736*w[0]**3*w[2]
 + 50.36082*w[0]**2*w[1]**2 + 51.23004*w[0]**2*w[1]*w[2] + 19.875*w[0]**2*w[2]**2
 + 38.88744*w[0]*w[1]**3 + 57.08256*w[0]*w[1]**2*w[2] + 40.94832*w[0]*w[1]*w[2]**2
     + 11.10984*w[0]*w[2]**3 + 13.61024*w[1]**4 + 26.82224*w[1]**3*w[2]
     + 31.76712*w[1]**2*w[2]**2 + 17.96152*w[1]*w[2]**3 + 9.70931*w[2]**4)

def objective(w):
   return mean(w)+var(w)+skew(w) + kurt(w)

For the constraint, in this case

def constraint1(w):
   return w[0]+w[1]+w[2]-1.0

for the initial guess, instead, let's use the portfolio that assigns the same weight
```

```
n = 3
w0 = np.zeros(n)
w0[0] = 0.333
w0[1] = 0.333
w0[2] = 0.333

And now we can complete the code.

# show initial objective
print('Initial Objective: ' + str(objective(w0)))

b = (0,10000)  #short selling is not allowed
bnds = (b, b, b)
con1 = {'type': 'eq', 'fun': constraint1}
cons = ([con1])
solution = minimize(objective,w0,method='SLSQP',
                    bounds=bnds,constraints=cons)
w = solution.x

# show final objective
print('Final Objective: ' + str(objective(w)))

# print solution
print('Solution')
print('w1 = ' + str(w[0]))
print('w2 = ' + str(w[1]))
print('w3 = ' + str(w[2]))
```

Finally, the solutions can be calculated and we find:

```
Initial Objective: 5.379044158345797
Final Objective: 4.229455361815727
```

Solution

```
w1 = 0.45074095500925276
w2 = 1.1058862159352145e-17
w3 = 0.5492590449907473
```

## 7. True Multicriteria Setups

In what we did, skewness and kurtosis were added to the optimization program and, by changing the coefficients $\alpha_i$s in the objective function we can decide whether mean, variance, skewness, etc, should be equally relevant. That approach works, but we saw that the calculations can become very cumbersome and basically untenable for portfolios with many assets.

In a 2010 paper, Mhiri and Prigent have proposed a different and more application oriented approach to portfolio optimization with higher moments. More specifically, we refer the section of their paper that concernns Portfolio optimization based on Polynomial Goal Programming(Section 2.2) and here we describe their method. If we let $\overline{R}, V, S, K$ denote the expected return vector, the co-variance matrix, the co-skewness matrix and the co-kurtosis matric, respectively, then we can run four individual optimization problems in which each optimization ignores the other goals, (i.e. we treat each problem independently of the other three):

$$\begin{aligned}&\max R^{\star}(w) = w^T\overline{R}\\&\text{subject to: } w^T\iota = 1\\&\qquad\qquad w_1 \geq 0 \ldots, w_m \geq 0;\end{aligned}$$

$$\begin{aligned}&\min V^{\star}(w) = w^T V w\\&\text{subject to: } w^T\iota = 1\\&\qquad\qquad w_1 \geq 0 \ldots, w_m \geq 0;\end{aligned}$$

$$\begin{aligned}&\max S^{\star}(w) = w^T S(w \otimes w)\\&\text{subject to: } w^T\iota = 1\\&\qquad\qquad w_1 \geq 0, \ldots, w_m \geq 0;\end{aligned}$$

and

$$\begin{aligned}&\min K^{\star}(w) = w^T K(w \otimes w \otimes w)\\&\text{subject to: } w^T\iota = 1\\&\qquad\qquad w_1 \geq 0, \ldots, w_m \geq 0.\end{aligned}$$

Let $R^{\star}, V^{\star}, S^{\star},$ and $K^{\star}$ be the optimal solutions for each of the 4 problems above. We label these values aspired levels, but clearly we cannot achieve them all at
the same time, not in general at least. Then, we call $d_1, d_2, d_3,$ and $d_4$ the goal variables which account for the deviations of expected return, variance, skewness and kurtosis from the respective aspired levels. Now, we run the following problem:

$$\begin{aligned}&\min Z = \left|\frac{d_1}{R^{\star}}\right|^{\lambda_1} + \left|\frac{d_2}{V^{\star}}\right|^{\lambda_2} + \left|\frac{d_3}{S^{\star}}\right|^{\lambda_3} + \left|\frac{d_4}{K^{\star}}\right|^{\lambda_4}\\&\text{subject to: } w^t\overline{R} + d_1 = R^{\star}\\&\qquad\qquad w^T V w - d_2 = V^{\star}\\&\qquad\qquad w^T S(w \otimes w) + d_3 = S^{\star}\\&\qquad\qquad w^T K(w \otimes w \otimes w) - d_4 = K^{\star}\\&\qquad\qquad w^T\iota = 1\\&\qquad\qquad w_1 \geq 0, \ldots, w_m \geq 0\\&\qquad\qquad d_1 \geq 0, d_2 \geq 0, d_3 \geq 0, d_4 \geq 0.\end{aligned}$$

Here $\lambda_1, \lambda_2, \lambda_3,$ and $\lambda_4$ represent investor preferences and are given before solving the problem. Hence they are parameters.The values
of the goal variables $d_i, i = 1, 2, 3, 4$ are supposed to measure the amount by which our selection process underachieves with respect to the best scenario for each problem.

Note that here we used the elegant matrix notation for skewness and kurtosis of the portfolio, but the paper proposes a different formula that avoid the Kronecker products and use instead
the sample definitions, but it is also necessarily much longer and much more cumbersome to write down. Practitioners will probably have to resort to use those, but here we wanted to stress
the methodology more than anything else.

There is one more way to approach the problem, one that amounts to a true multicriteria decision analysis (MCDA). This is more complicated and it leads to multiobjective programming in
which more than one objective functions are optimized simultaneously. A multiobjective optimization looks like this:

$$\begin{aligned}&\min_{w} \{f_1(w), f_2(w), \ldots, f_r(w)\}\\&\text{subject to } \; Aw \leq a\\&\qquad\qquad w \geq 0\end{aligned}$$

where $f_1(w) = w^T M_1$, $f_2(w) = w^T M_2 w,$ etc. In multiobjective programming, we do not have an optimal solution as it is unfeasible to optimize
all objective functions simultaneously. For this reason, the concept of optimal solution is replaced by the concept of {\bf Pareto optimal} solution.
A feasible solution $w$ has the property to be {\bf Pareto dominant}, i.e., if $v$ is another feasible solution, then $f_i(w) \leq f_i(v)$ for $i = 1, 2, \ldots, r$ and $f_i(w) < f_i(v)$ for at least one value of $i = 1, \ldots, r.$

## 8. Conclusion

This lesson introduced a more general way to perform portfolio selection by taking into consideration skewness and kurtosis of the resulting portfolios directly. In addition, it also showed how to account for liquidity. Finally, we discussed an example of a

true multi-objective optimization problem, which is a much more difficult problem, and we will see a bit more of it in Module 3, Lesson 4.

In the next lesson, we will study how to deal with currency management, an issue that is of importance when a portfolio does not consist of assets all denominated in the same domestic currency.

**References**


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